APPENDIX C

Table 2- Categories and performance qualification levels of water quality for medical device processing.

Water Quality Measurement	Units	Utility Water	Critical Water	Steam*
pH @ 25 °C:	pH	6.5 – 9.5	5.0 – 7.5	5.0 – 9.2**
Total Alkalinity	mg $CaCO_3$/L	<400	<8	<8
Bacteria	CFU/mL	<500***	<10	N/A
Endotoxin	EU/mL	N/A***	<10	N/A
Total Organic Carbon (TOC)	mg/L (ppm)	N/A	<1.0	N/A
Color and Turbidity	Visual	Colorless, clear, without sediment	Colorless, clear, without sediment	Colorless, clear, without sediment
Ionic Contaminants				
Aluminum	mg/L	<0.1	<0.1	<0.1
Chloride	mg/L	<250	<1	<1
Conductivity	µS/cm	<500	<10	<10
Copper	mg/L	<0.1	<0.1	<0.1
Iron	mg/L	<0.1	<0.1	<0.1
Manganese	mg/L	<0.1	<0.1	<0.1
Nitrate	mg/L	<10	<1	<1
Phosphate	mg/L	<5	<1	<1
Sulfate	mg/L	<150	<1	<1
Silicate	mg/L	<50	<1	<1
Total Hardness	mg $CaCO_3$/L	<150****	<1	<1
Zinc	mg/L	<0.1	<0.1	<0.1

Rationale for Table 2 water quality parameters and other considerations:

a) **pH:** Water having extremes in pH can cause damage of device surfaces. In addition, an extreme pH can interfere with the efficacy of cleaning agents and disinfectants (see E.2.18). Utility water is expected to have a pH in the range of 6.5 to 9.5. The pH of the water will be dependent on the disinfection chemistry (e.g., chlorine, monochloramine) and water treatment strategy of the local municipality. Potable water is typically in the higher portion of this range. For Critical Water the specification is based upon typical values for the water produced. For boiler-treated steam that travels through an extensive distribution network that can be oxidized (e.g., black iron), acidic pH can result in corrosion of the piping and transfer of rust to the sterilizer load, therefore pH should be controlled to maintain a pH of 7.5 to 9.2. For steam produced by a dedicated generator the pH can be in the range of 5.0 to 7.5 (see EN 285).

b) **Alkalinity:** Total alkalinity is the measure of dissolved solids which are alkaline in nature such as bicarbonate, carbonate, and hydroxide. High alkalinity levels increase the tendency of hard water salts to form scale. High alkalinity in the final rinse water increases potential for concentration of alkaline dissolved solids onto the device during drying. This dried alkaline residue can then be reactivated under the high temperature and humidity conditions of a steam sterilizer causing devices to become stained or corroded. The performance of cleaning agents that are not buffered may be impacted by excess alkalinity. Different types of alkalinity may buffer differently and impact scale type or cleaning agents in different ways.

c) **Bacteria:** Critical Water should have very low levels of bacteria. Although water treatment equipment may produce water that is essentially free from microorganisms, the water distribution system may allow for introduction of low levels of bacteria, requiring routine monitoring. Levels higher than the bacterial levels in Table 2 may increase the bioburden on the load and can increase the risk of patient infections and pyrogenic reaction due to increased endotoxin levels generated by certain types of bacteria.

d) **Endotoxin:** The limit specified by United States Pharmacopeia (USP) for transfusion and infusion assemblies and similar medical devices (USP <161>) is 20 EU remaining on its surface. Critical Water can be derived by a process that would be expected to remove endotoxin. While treatment can reduce endotoxin, some endotoxin might be detected because of passage of the water through tubing that is not endotoxin-free. If the final rinse of a medical device is done with Critical Water containing less than 10 EU/mL, the amount of endotoxin left on the device would be expected to be only a fraction of that amount so the device would have far less than the allowed limit of 20 EU remaining on its surface. Because Utility Water is not produced by a process that removes endotoxin and it is not used for the final rinse of critical devices that contact the patient's bloodstream, endotoxin testing of such water is unnecessary unless it is used as the rinse water after high-level disinfection of medical devices because this rinsing is after the antimicrobial process.

NOTE Adverse effects have been demonstrated when 1 to 4 ng/kg of purified endotoxin (1 ng/kg equals 2 EU/kg) are injected [86]. However, the threshold amount of endotoxin in rinse water that would lead to sufficient residuals on a reprocessed medical device to cause an adverse patient reaction is not clearly defined in the literature. There are no published data or guidelines for endotoxin levels in water used to reprocess medical devices. The 10 EU/mL value described here was adopted by AAMI WG95 committee consensus on the grounds that it was thought unlikely to result in sufficient residual endotoxin on a device to cause an adverse patient reaction.

e) **Total organic carbon (TOC):** Residual TOC should be minimized due to the unknown nature of the TOC and its potential effect on patient tissue and mucosa. The TOC value for Critical Water is based on the expected efficacy of carbon filtration, RO, or distillation in removing organic material and has been defined as less than 1.0 mg/L (ppm). Utility Water would be expected to have total organic carbon levels that are not significantly different than tap or untreated water. Water with high levels of TOC can impact efficacy of UV lamps at controlling microbial growth in water storage systems.

NOTE No studies have been published that establish how much organic material is needed to interfere with disinfection and sterilization processes.

f) **Color and turbidity:** EPA drinking water standards require water to be colorless and clear and contain no sediment. Color and residues are indicative of contaminants and increase the potential to leave contaminants on medical devices during processing. Contaminants left behind on medical devices during certain parts of the processing steps can both stain and/or interfere with subsequent processes. The same criteria for color have been applied to both categories.

6.3.1 Ionic characteristics

a) **Conductivity:** this is a measure of the total amount of electrically charged impurities (ions) that are present in the water sample. It is measured in terms of electrical current over a distance with units of µS/cm. It is a broad measurement of the impurities in the sample and is used as a proxy for Total Dissolved Solids. It gives an indication of the general purity of the sample. High conductivity indicates the presence of large amounts of charged ions, some of which can result in corrosion or staining of instruments.

b) **Chloride:** When chloride ions are present in sufficient concentrations they can result in corrosion of stainless steel or soft metals.

c) **Nitrate:** residual nitrates can be very reactive with other compounds. Residual amounts of nitrates on medical device surfaces may affect subsequent processes.

d) **Phosphate:** phosphates can enter water as run off from agricultural uses and can contribute to scale formation, leave residues on instrumentation, and interact with cleaning chemicals

Sulfate: sulfates can form scale with positively charged ions. Routinely monitored water treatment systems with increasing levels of sulfate can be indicative of resin bed issues.

f)**Silicate**: silicates can form hard to remove residues, especially if allowed to remain on instruments during subsequent steps of processing involving heat.

g)**Iron**: iron ions can deposit on stainless steel surfaces and cause corrosion. It can also result in rust/corrosion that can manifest in a variety of colors (e.g., brown, black, etc.).

h)**Copper**: copper ions can deposit on surfaces of items being processed and cause green staining and/or corrosion.

i)**Manganese**: manganese ions can deposit on surfaces of items being processed and cause black staining and/or corrosion.

j)**Aluminum**: aluminum ions can deposit on surfaces of items being processed and cause staining and/or corrosion.

k)**Zinc**: zinc ions can deposit on surfaces of items being processed and cause staining and/or corrosion.

NOTE Metal ions (e.g., iron, copper, zinc, etc.) can also contribute to hardness values and deposit on metal surfaces resulting in galvanic corrosion.

l)**Water hardness**: Water hardness is a critical parameter to control as it affects residues on devices. Tap water is generally considered "hard" if it has calcium and magnesium levels higher than 150 ppm; therefore, this level is given as the upper limit for Utility Water. Avoiding excess hardness prolongs the life of washer-disinfectors; reduces the risk of hard-water deposits on medical devices during processing and helps ensure the efficacy of cleaning agents. If hardness is greater than 150 mg/L(ppm), water softening is recommended unless used for washing where the cleaning chemistry is capable of handling higher levels of hardness or metal ions in solution. For Critical Water, the treatment process would be expected to remove contaminants and produce water that has calcium and magnesium levels of less than 1 ppm. Using Critical Water for the final rinse of critical medical devices reduces the likelihood of hardwater deposits.

There are performance qualification specifications for water quality defined in Table 2. For ongoing monitoring of the quality, a reduced set of specifications with greater frequency (Table 4), allows for setting of a baseline of expected quality parameters. The values expressed in Table 4 (i.e., action levels) are not to be exceeded, however trending baseline data and setting an alert level can give early indication of the water quality trend. The baseline data can be used for indicating variations that may impact the water quality and setting of appropriate alert levels.

The alert levels can be identified using a method determined by the facility's multidisciplinary team or can be calculated using the example provided in Clause G.1.2.

An alert should be triggered if a sample results in data outside of the range defined by the upper control limit (UCL) and the lower control limit (LCL).

When data is outside the upper or lower alert levels, it is an excursion, a deviation from a regular pattern, path, or level of operation. Any excursions outside of those typically seen values (within action levels) can be identified as warning of potential issues. The monitoring program is an opportunity to address potential problematic issues prior to failing of specifications and can help anticipate repair or replacement of water treatment equipment or processes.

An excursion (e.g., data point outside alert level) should prompt investigation in order to find the assignable causes and associated risks and then mitigate, where possible. These causes may indicate a change in conditions (e.g., seasonal, construction, repair).

11 Continuous quality improvement

11.1 Introduction

This section identifies performance measures and process monitors that can be used for continuous quality improvement (CQI) programs. CQI programs are recognized as an effective means of improving the performance of any process. For water quality, a CQI program encompasses the entire process of decontamination through sterilization. The quality of the water is important in each stage and needs to be monitored at various times in the process.

11.2 Quality process

Procedures for determining water quality should be based on a documented quality process that measures objective performance criteria (e.g., pH, hardness, purity, temperature) that have a direct effect on the outcome. This quality process should be developed in conjunction with personnel from other areas (e.g., facility engineering, the sterile processing department, infection prevention and control) and integrated into the overall quality process in the health care facility. Variables in the system may be controlled to achieve assurance of product quality and process efficacy. Monitoring frequency will vary, depending on the quality improvement goals, the health care facility policies and procedures for the handling of unfavorable or unplanned events, and the type of process variable.

An investigation should be conducted for any problem relating to any aspect of water quality. Problems and updates should be reported to those involved with medical device processing.

There should be a planned, systematic, and ongoing process for verifying compliance with procedures. Quality processes can be enhanced by audits that are conducted on a regular basis. The information from these activities should be summarized and made available to individuals, groups, or teams as needed.

Table 5 and Table 6 describe the water quality monitoring that shall be performed as part of a quality improvement process within their area. Continuous quality improvement involves conducting risk analyses.

Water treatment systems maintenance

12.1 General considerations

Proper maintenance of a well-designed water treatment system should follow the original equipment manufacturers (OEM) recommendations. Improper maintenance can lead to equipment performance issues, water system component failure, and/or improper water quality levels that do not meet the guidelines of this standard. Bacterial growth, increased endotoxin levels, and poor final water quality are indicative of a poorly maintained water treatment and delivery system.

Loop systems and tanks (see Annex G.4) should be disinfected monthly as a best practice.

A regular system maintenance schedule should be followed to meet OEM recommendations including expendable components based on site specific water conditions, at a minimum, and indicators should be defined for each piece of equipment in the water treatment system.

It is recommended that a daily checklist be used to monitor basic system parameters, such as loop final water quality and critical OEM directed readings, which may include levels of expendable items that need to be monitored.

Testing of at least the loop out and the loop return should be done no less frequently than monthly to verify bacterial levels and endotoxin levels.

12.2 Serviceability of system

Systems shall be designed with clear guidance on system controls that are easy to follow and reduce errors. The system should be allotted adequate room for installation that will allow access to all components for maintenance and allow adequate access around the equipment. Space should be allowed in between components and the ends of the system to allow for replacement of expendable items and repairs. Systems should meet all local building codes.

Reverse osmosis systems should be set up to permit disinfection and allow for low and high pH chemical cleanings for ongoing maintenance.

Systems should be on a fully recirculating loop to allow for full and effective disinfection of the system piping and have a fully drainable tank for complete and faster disinfection. Existing direct feed systems should be updated to recirculating systems, when possible, to allow for effective disinfection. Until such time that the loop is installed, the direct feed may be required to be monitored and disinfected more frequently to reduce the risk of bacterial growth and/or endotoxin accumulation.

All newly designed and installed systems shall be on a fully recirculating loop.

Sample ports shall be placed at the loop out after final filtration as well as right before re-entry into the storage tank. Where feasible, an accessible sample port should be placed before individual points of use (e.g., immediately before the inlet to the washer-disinfector).

13 Special considerations

13.1 Post construction or extended shutdown

Typically performed by facilities engineering personnel:

- Reverse osmosis system/ultra filtration should be flushed. Flushing should be run every 24 hours for a minimum of 30 minutes or follow OEM recommendations.

- Pretreatment apparatus or filters should be scheduled to backwash or regenerate every 48-hours maximum.

- Pretreatment filter(s) should be changed out prior to putting into service.

- Water treatment equipment should be disinfected prior to putting into service.

- Flush Critical Water systems, by running water for two system volumes post disinfection. Contact the processing equipment manufacturer for guidance regarding how to flush stagnant water from equipment.

- Remove aerators and flush hot and cold sink water for a minimum of 5-minutes or until visibly clear and free of particles/debris.

- Total bacteria and endotoxin tests should be performed prior to putting into service.

- After construction, or an extended period of disuse, Critical Water systems should be flushed to remove stagnant water prior to sampling. Bacterial testing and endotoxin testing should be performed on the loop.

- Upon receipt of total bacteria and endotoxin tests with acceptable results, the unit may be placed into service.

13.2 Extended boil water alerts and steps to take after alerts are lifted

- Once the boil water alert has been lifted, flush all equipment (e.g., water treatment equipment/loop) with a disinfectant that is compatible with piping materials and equipment and restart. Also, flush feedwater source (prior to treatment equipment). o After disinfectant application is complete, flush for at least five minutes and until residual disinfectant is detected at feedwater source prior to treatment.

o If chlorine cannot be detected at point-of-use after flushing, supplemental disinfection may need to be considered (drinking water regulations will apply, by performing supplement disinfection).

- Drain, disinfect, flush, and refill water storage tank.
- Change pretreatment filters, backwash carbon tanks, regenerate softener, clean and disinfect RO membranes, disinfect distribution loop.
- If steam sterilizers are used in processing, check the steam system.
    - Check water quality used for steam generation.
    - Check with manufacturer for written instructions for bringing the system back online following a contamination event.

- If using deionization tanks these should be replaced with new deionization tanks prior to going back into service.

- The loop final filter should be replaced with a new absolute rated final filter.

13.3 Interruptions in service

If an interruption of service is from the utility feedwater source or a building system, shut down and restart.

- Apply disinfectant and flush for at least five minutes until residual disinfectant is detected at feedwater source prior to treatment. Disconnect the feed to the unit and flush the feed source directly to the drain.

- If disinfectant cannot be detected at point-of-use after flushing one may need to consider supplemental disinfection (drinking water regulations will apply, by performing supplemental disinfection the health care facility would now be considered a small public water system).

If the system cannot be disconnected from the feed source and directed to drain:
 Change pretreatment filters, backwash carbon tanks, regenerate softener, clean and disinfect RO membranes, and disinfect distribution loop.

 If steam sterilizers are used in processing, check the steam system. o Check water quality used for steam generation.

o Check with manufacturer for instructions for bringing system back online following a contamination event.

 If using deionization tanks these should be replaced with new deionization tanks prior to going back into service.
 The loop final filter should be replaced with a new Absolute Rated Final Filter.

13.4 System repair, modification and/or routine maintenance on a Critical Water production, storage, and distribution system

A Critical Water system is or should be designed to deliver water meeting all specified parameters to the points of use. See Clause 8.3 for details of Critical Water system design and Table 2 for the specified parameters.

Routine maintenance, repair, or system expansion can require that the Critical Water system be opened to the atmosphere, which exposes the piping and water in the system to ambient airborne microbiological flora. In the interests of patient safety, these should be assumed to be pathogenic. Thus, any time piping is replaced, or system modification is completed on a Critical Water system, the system piping and storage should be disinfected.

Disinfection should be performed after installation, expansion, major repairs, non-compliant bacterial levels, or other changes to the system (e.g., nearby building construction or water supply interruptions). In addition, disinfection should ideally be completed for prophylactic purposes, even if no non-compliant bacterial or endotoxin results have been observed (See Annex G).

Specific instances in normal maintenance in which system disinfection is required include:

13.4.1 Pretreatment equipment

Examples: cartridge particulate filtration, multimedia filtration, softener, carbon filter, sodium bisulfite injection system, and anti-scalant injection system

This equipment typically does not require regular disinfection. However, if testing indicates non-compliant bacterial levels at the sample port immediately downstream of the pretreatment equipment, the equipment should be disinfected according to material compatibility. If testing still indicates non-compliant bacterial levels, the filtration media should be replaced. Non-compliant levels are defined here as a bioburden level of too numerous to count.

13.4.2 Primary treatment equipment

EXAMPLE: reverse osmosis (RO), deionization (DI), and electrodeionization (EDI)

13.4.2.1 Reverse Osmosis (RO) unit

RO units should be cleaned, using high and low pH cleaning chemicals, when permeate (product water) indicates membrane fouling and degradation of performance. This will be dependent upon the quality of the feedwater to the RO unit, as well as the volume of water filtered by the RO. Therefore, frequency should be determined by the specific use profile of the sterile processing department and when data indicates cleaning is required. At a minimum, this should be performed at least annually.

RO units should also be disinfected using a compatible disinfection method at least annually or if testing indicates non-compliant bacterial levels at the sample port immediately downstream of the RO unit.

13.4.2.2 Deionization (DI) exchange tanks

DI exchange tanks are typically not compatible with the disinfection methods. Therefore, the DI exchange tanks should be exchanged with replacement tanks when testing indicates non-compliant bacterial levels at the sample port immediately downstream of the DI tanks and there is not a bacteriostatic filter downstream of the DI tanks.

13.4.2.3 Electrode ionization (EDI) unit

EDI units should be disinfected when testing indicates non-compliant bacterial levels at the sample port immediately downstream of the EDI unit.

13.4.3 Storage tanks and distribution equipment and piping

Examples: storage tanks, distribution pump(s), UV disinfector, bacteriostatic filtration, and distribution loop piping
High purity water storage, distribution equipment and piping should be disinfected using a compatible disinfection method if testing indicates non-compliant bacterial levels at the sample port immediately downstream of the final treatment technology, typically the bacteriostatic filters, at the inlet of the distribution loop, typically the bacteriostatic filters and on the sample port returning. If bacteriostatic filters are not compatible with disinfection methods (see Annex G), the filters must be replaced with new filters. Bacteriostatic filters should be replaced if the pressure difference (delta pressure, or ΔP) across the filter is 10 PSI or greater, or downstream testing indicates biological breakthrough and/or loop biofouling. Localized disinfection is required for areas that can be bypassed for normal maintenance. For example, the bacteriostatic filter if there is a bypass permitting its replacement without exposing the entire water system to ambient atmospheric conditions. Bypasses should be designed to ensure that stagnant water within them does not allow for bacterial growth or biofilm formation. If the bacteriostatic filter at the inlet of the distribution loop cannot be isolated during replacement, the distribution loop should be disinfected after the filters are replaced.
Storage tanks, distribution equipment, and distribution loop piping should be disinfected after the replacement of in-line probes and sensors. Storage tanks, distribution equipment, and distribution loop piping should be disinfected after service and/or replacement of recirculation pumps and associated piping. Storage tanks, distribution equipment, and distribution loop piping should be disinfected after a shutdown of a maximum duration determined by the multidisciplinary water management team, or when post-shutdown testing, if performed, indicates non-compliant bacterial levels.

Microbiological quality – bacterial endotoxin test

H.4.1 *Limulus* amebocyte lysate (LAL)

Testing for bacterial endotoxin is generally performed by means of the *Limulus* amebocyte lysate (LAL) assay. Two basic types of LAL assay can be performed: the kinetic assay, which is available in a colorimetric or turbidimetric format, and the gel-clot assay. Alternatively, validated test methods to detect endotoxin or pyrogens may be used. The LAL assay is described in ANSI/AAMI ST72 [7]. A positive control and a negative control should be run with each assay. The limits for endotoxin levels are given in Table 2.

How to perform water sampling

H.5.1 Sample collection

Critical Water samples, or other types of water when indicated, should be collected from several locations within the distribution system to provide a representative microbial water quality of the system. In general, samples should be collected at the beginning and end of the water distribution loop. A point of use in the distribution loop is a location where water enters equipment utilized to process medical devices. Where any storage tank is used, water should be sampled at the end of the purification process and the exit of the storage tank.
Samples should always be collected before disinfection of the water treatment system. If repeat cultures are performed after the system has been chemically disinfected (e.g., with hydrogen peroxide, chlorine, or peracetic acid), the system should be flushed completely before samples are collected. Before samples are collected, the storage tanks and distribution system should be drained and flushed until residual disinfectant is no longer detected. For new systems, the water should be cultured weekly until an established pattern has been determined. For established systems, the water can be cultured monthly unless a greater frequency is dictated by historical data at a given institution.
Water samples should be collected directly from outlet taps situated in different parts of the water distribution system. In general, the sample taps should be opened, and the water allowed to run for at least 60 seconds before a sample is aseptically collected in a sterile, endotoxin-free container. The collection container should not be made of polypropylene, because polypropylene has been shown to inhibit endotoxin detection [78]. At least 100 to 300 mL of water should be collected. Sample taps should be treated with a general disinfectant (e.g., alcohol) prior to sampling and be completely dried to prevent contamination of the collected sample disinfectant

residual from the sample tap. Bleach or other disinfectant solutions should never be used. Bacterial replication is prevented by holding the sample on ice, or at 4 °C, until it is delivered to the laboratory for testing.

H.5.2 Water sampling technique

Sampling Consistency: Sampling should be consistent in procedure and sampling locations to ensure the accurate monitoring of water quality.

Flushing: Sample points should be flushed commensurate to the size of the water system and distribution loop. Biofilms can develop throughout the water system, including infrequently used valves or downstream connectors between the distribution loop and equipment. Flushing these areas with a high flow velocity across these surfaces can shear off fragile tops of biofilms possibility growing in these areas. A standardized flush before sampling can limit the microbial contribution from the sampling points.

Sampling containers: The appropriate sampling container should be used to collect the water sample. The container should be of the required quality not to introduce contamination into the test sample. For example, samples used in the sampling for bacterial endotoxin should be of the appropriate quality (e.g., pyrogen-free).

Sampling technique: Valves/taps/hoses used for sampling should be of the appropriate design to not contaminate the sample. For example, sampling locations do not provide a stagnant water environment when not in use and is accessible to the distribution loop disinfection process. Prior to sampling the sampling surfaces (e.g., valves, hoses) should be disinfected both externally and internally to reduce microbial contamination. Aseptic technique should be used for water collection.

Results exceeding specification

Microbial quality levels detected, and exceeding specification should lead to a repetition of the disinfection and sampling at the location to confirm the results. Upon confirmation of the exceedance, immediate action is required to reduce the microbial contamination. The distribution loop should be disinfected using methods described in Annex E to ensure the microbial contamination is not being introduced through the distribution system. Automated instrument reprocessors (e.g., washer-disinfectors, washer-decontaminators, AERs) should be flushed with 1,000 ppm of a chlorine-releasing agent after the first instance of persistent unacceptable microbial levels is detected [88]. Other agents that could be considered for flushing include chlorine dioxide, peracetic acid, hydrogen peroxide, and super oxidized water. Before the unit is flushed with any of these agents, it is important to confirm with the reprocessor manufacturer that the components of the machine are compatible with the disinfectant to be used in its intended concentration. The levels should be monitored weekly by repeat testing until two successive counts show that baseline levels have been achieved.

If the microbial levels continue to exceed acceptable limits despite disinfection activities, a second system disinfection, combined with replacement of as much of the pipework as is practical, should be undertaken to remove biofilm. For the second round of disinfection, a higher concentration of chlorine-releasing agent (e.g., 10,000 ppm) or an alternative disinfectant should be used [88].

If contamination problems persist despite all efforts, the main water source should be investigated, and local authorities asked for input. This approach is based on the action plan outlined by the Working Party of the Hospital Infection Society and the Public Health Laboratory Service [88].